

## Handout 02.11: Interest Math Practice

Name: \_\_\_\_\_ Date: \_\_\_\_\_ Partner: \_\_\_\_\_

Rates on this sheet are typical for 2026 and to be refreshed before teaching. The people are made up.

**My do now guess:** a \$500 phone at \$25 a month on a store card will cost \$\_\_\_\_\_ in total.

### Part A: Debit vs credit, side by side

|                               | Debit card | Credit card |
|-------------------------------|------------|-------------|
| Whose money is it?            |            |             |
| When does it leave?           |            |             |
| Is there a bill?              |            |             |
| Is there interest?            |            |             |
| What if the card is stolen?   |            |             |
| Does it build a credit score? |            |             |

**Credit** is \_\_\_\_\_

**Debt** is \_\_\_\_\_

**Interest** is \_\_\_\_\_

### Part B: Simple interest on \$500

**The formula:**  $I = P \times r \times t$  (interest = amount borrowed x rate as a decimal x years)

Rates as decimals: 5 percent = 0.05. 15 percent = 0.15. 24 percent = 0.24.

Worked example: \$500 at 5 percent for 1 year.  $\$500 \times 0.05 \times 1 = \$25$ . Total owed:  $\$500 + \$25 = \$525$ .

| Rate       | What it is like       | One year: interest                      | One year: total owed | Three years: interest                   | Three years: total owed |
|------------|-----------------------|-----------------------------------------|----------------------|-----------------------------------------|-------------------------|
| 5 percent  | a good car loan       | $\$500 \times 0.05 \times 1 = \$$ _____ | \$_____              | $\$500 \times 0.05 \times 3 = \$$ _____ | \$_____                 |
| 15 percent | a credit union card   | $\$500 \times 0.15 \times 1 = \$$ _____ | \$_____              | $\$500 \times 0.15 \times 3 = \$$ _____ | \$_____                 |
| 24 percent | a typical bank credit | $\$500 \times 0.24 \times 1 = \$$ _____ | \$_____              | $\$500 \times 0.24 \times 3 = \$$ _____ | \$_____                 |

|  |      |  |  |  |  |
|--|------|--|--|--|--|
|  | card |  |  |  |  |
|--|------|--|--|--|--|

At which rate does three years of interest cost more than half the phone? \_\_\_\_\_

(Grade 6: one-year columns only.)

### Part C: Two more loans

1. Ines borrows \$1,200 on a store card at 15 percent and pays nothing for two years.  
Interest: \$\_\_\_\_\_ x \_\_\_\_\_ x \_\_\_\_\_ = \$\_\_\_\_\_. Total owed: \$\_\_\_\_\_.
2. A used car loan: \$8,000 at 5 percent for four years. Interest: \$\_\_\_\_\_ x \_\_\_\_\_ x \_\_\_\_\_ = \$\_\_\_\_\_. Total paid: \$\_\_\_\_\_.
3. Which loan cost more in dollars? \_\_\_\_\_ Which cost more as a percent of what was borrowed? \_\_\_\_\_ How can both be true?

\_\_\_\_\_

### Part D: The minimum payment trap

Leo puts a \$500 phone on a credit card at 24 percent APR. That is 2 percent a month. Each month the card adds 2 percent of the balance, then his payment comes off.

The first three months at \$25 a month (follow along on the board):

| Month | Balance at start | Add 2 percent | Pay \$25 | Balance at end |
|-------|------------------|---------------|----------|----------------|
| 1     | \$500.00         | \$10.00       | \$25.00  | \$485.00       |
| 2     | \$485.00         | \$9.70        | \$25.00  | \$469.70       |
| 3     | \$469.70         | \$9.39        | \$25.00  | \$454.09       |

Three months in, Leo has paid \$75 and still owes \$454.09. Copy the finished results from the slide:

| Plan                         | Months to pay off | Total interest | Total paid for the \$500 phone |
|------------------------------|-------------------|----------------|--------------------------------|
| \$25 a month (the minimum)   |                   |                |                                |
| \$50 a month                 |                   |                |                                |
| Paid in full the first month |                   |                |                                |

**My recommendation (two sentences, with the numbers):**

\_\_\_\_\_

\_\_\_\_\_

My do now guess was \$ \_\_\_\_\_. The real number was \$ \_\_\_\_\_. I was off by \$ \_\_\_\_\_.

### Part E: The credit score, in one paragraph

A credit score is a number from \_\_\_\_\_ to \_\_\_\_\_ that shows \_\_\_\_\_.

It goes up when I \_\_\_\_\_ and \_\_\_\_\_.

It goes down when I \_\_\_\_\_ or \_\_\_\_\_.

It matters because \_\_\_\_\_, \_\_\_\_\_, and sometimes \_\_\_\_\_ look at it.

### Grade 8 stretch

1. Do months 4 and 5 of the \$25 plan by hand and check them against the pattern.
2. What monthly payment would clear the \$500 in ten months? (Try \$55, then \$56.) About how much interest is that?
3. Three cards: a bank card at 24 percent, a store card at 29 percent, a credit union card at 15 percent. A person with a good score should pick \_\_\_\_\_ because \_\_\_\_\_.

### Exit card (3-2-1)

3 differences between credit and debit: \_\_\_\_\_, \_\_\_\_\_, \_\_\_\_\_

2 things that move a credit score: \_\_\_\_\_, \_\_\_\_\_

1 number: the total cost of the \$500 phone at \$25 a month: \$ \_\_\_\_\_